

Conducting Laboratory

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Intended Learning Outcomes

- To be aware of the role of a lab tutor
- To identify the importance of teaching and learning activities (TLAs) and assessment tasks in a laboratory session
- To recognize specific strategies for handling laboratory class

Why do students need laboratory sessions?

- Learning by doing, make abstract concepts concrete
- Gain practical, hands-on experience
- Verify theories
- Learn scientific methods and techniques
- Foster curiosity and creativity
- Develop critical thinking skills and communication skills

Role and Responsibilities of a Lab Tutor

- Key attributes
 - Approachable, knowledgeable, fair, passionate, clear-speaking
- Responsibilities
 - Prepare and plan the laboratory session
 - Ensure safety
 - Demonstrate experiments, guide experiment execution, handle challenges
 - Provide feedback, grade assessments

Preparation and Planning

- The laboratory
 - Safety
 - Lab technicians
 - Emergency contact (security hotline 3442 8888)
- The experiment
 - Lab manual, rehearsal, common challenges
- The course
 - Syllabus and ILOs
 - Assessment criteria

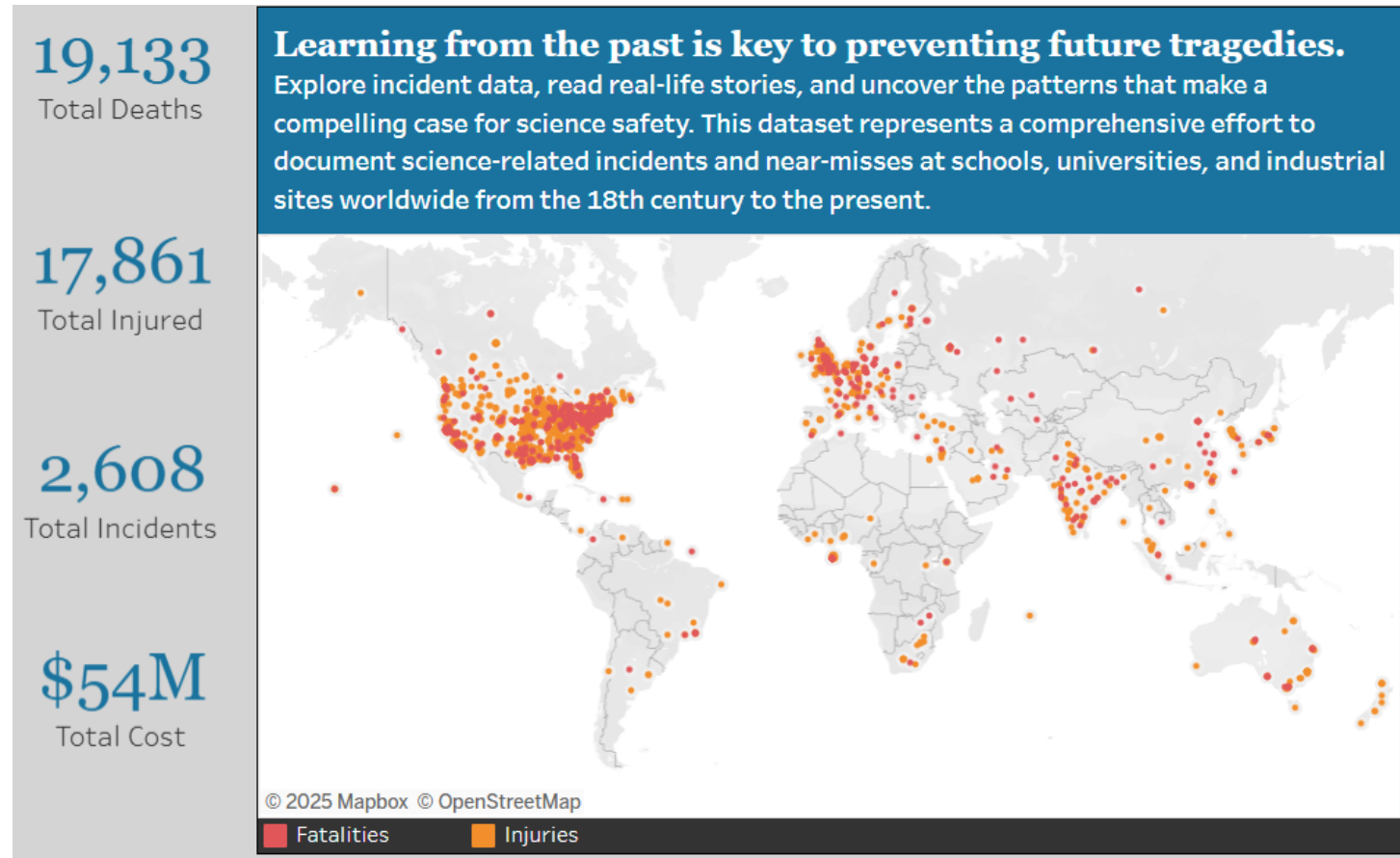
History of Science Incidents

Indrajit Basak (2025)

National Institute of Technology, India

“Ignition of aluminium powder in a furnace at an outdoor demonstration of 'Thermit welding' for final-year mechanical engineering students may have produced hydrogen gas unexpectedly, leading to an explosion. Professor Basak, due to his proximity to the furnace, sustained serious burns and died a week later.”

<https://www.labsafety.org/memorial-wall>
<https://www.labsafety.org/science-incident-dashboard>



(Start from 1760)

Safety Management

- Legislation: Occupational Safety and Health Ordinance, Chapter 509
- CityUHK Online Safety Training: Provided by the Facilities Management Office (FMO) on Canvas
 - Ensure all students have completed the required safety training set by the College/School/Department before conducting laboratory activities
 - General Safety Training: Safety, Health and Environment Protection @ CityU
 - Laboratory Safety Training:
 - Fundamentals (including **emergency procedures**); Policy; Risk; Waste management
 - Chemical; Biological; Animal; Radiation; Laser; Electrical and Mechanical

Symbols for Hazard Warnings Protections Required

1		Biological Hazard 生物危害	2		Chemicals Reactive with Water 遇濕易燃化學品	3		Corrosive Chemicals 腐蝕性化學品		
4		Cryogenic Liquid 極低溫液體	5		Electrical Hazard 電力危險	6		Flammable Chemicals 易燃化學品		
7		Gas Cylinders 壓縮氣體	8		High Temperature 高溫	9		High Voltage 高壓電力		
10	Humidity Control Zone 濕度控制區		11		Ionizing Radiation 電離輻射	12		Laser Radiation 激光		
13		Low Temperature 低溫	14		Non-ionizing Radiation 非電離輻射	15		Oxidizing Chemicals 助燃化學品		
16		Slippery Floor 小心地滑	17		Strong Magnetic Field 強力磁場	18		Toxic Chemicals 有毒化學品		
19		Ultra-violet Light 紫外光								
			1		NO ACCESS for person with metal implants 用金屬義肢者， 不得內進	2		NO ACCESS for person with pacemaker 用心臟起搏器者， 不得內進	3	

1		NO ACCESS for person with metal implants 用金屬義肢者，不得內進	2		NO ACCESS for person with pacemaker 用心臟起搏器者，不得內進	3		Wear Dust Mask 佩戴防塵口罩
4		Wear Ear Protection 佩戴護耳罩	5		Wear Eye Protection 佩戴護眼罩	6		Wear Face Shield 佩戴護面罩
7		Wear Gloves 穿戴手套	8		Wear Helmet 佩戴頭盔	9		Wear Laboratory Gown 穿著實驗室袍
10		Wear Laser Goggles 佩戴激光護眼罩	11		Wear Safety Shoes 穿著安全鞋	12		Wear Welding Mask 佩戴燒焊面罩

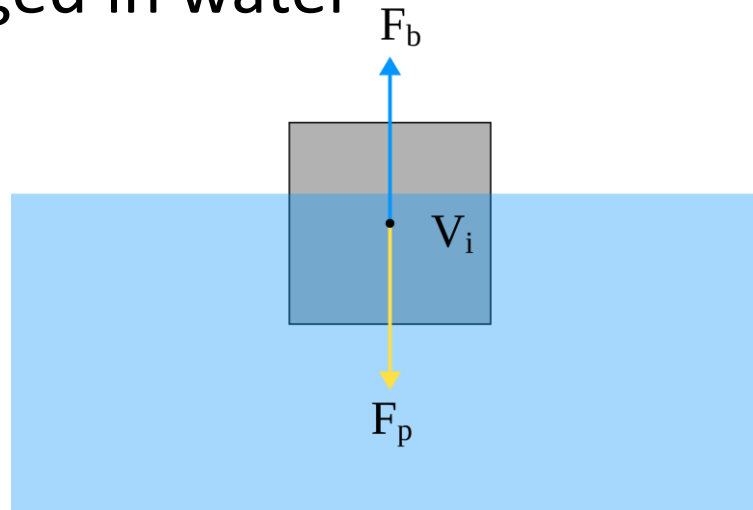
Adopted from “Guidelines for Preparing and Displaying the Laboratory Door Safety Signage” by the Facilities Management Office

Common questions that we should ask teachers during the preparation

- "Are there any pre-lab activities or readings I should review or assign to students?"
- "What key concepts or skills should students master, and how can I reinforce them during the session?"
- "How does this lab connect to lecture material or broader course goals?"
- "How should I assess student work?"
- "What feedback should I provide to students, and how do I report grades or issues?"
- "Is there a plan for handling equipment failures or incomplete sessions?"

Demonstrate an experiment: Case Study

- Theory: Archimedes' Principle
 - *"Any object, totally or partially immersed in a fluid or liquid, is buoyed up by a force equal to the weight of the fluid displaced by the object."*
Floating Bodies, Archimedes (c. 246 BC)
- Goal: To verify Archimedes' Principle by measuring the buoyant force on an object submerged in water



$$F_b = \rho_{\text{fluid}} \times g \times V_{\text{displaced}}$$

$$\text{Density} = \frac{\text{Mass}}{\text{Volume}}$$

- Apparatus, Safety Precautions

- Procedure

1. Measure weight in air

1. Suspend the metal cylinder from the spring balance
2. Record its weight in air, W_{air}

2. Prepare overflow can with water

1. Fill the overflow can with water until it just begins to drip from the spout
2. Wait until dripping stops

3. Submerge the object

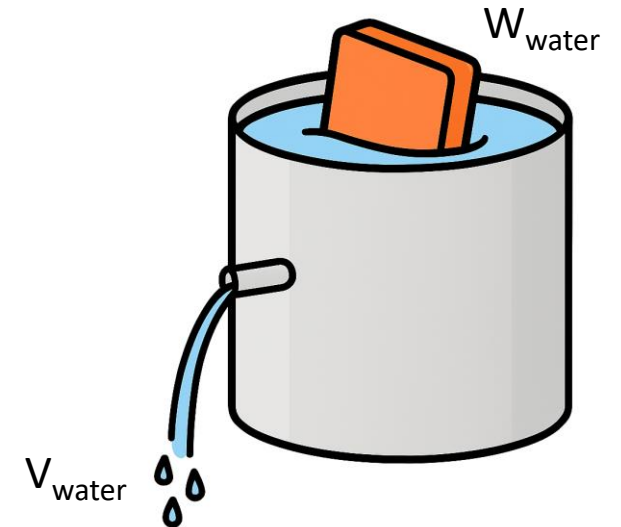
1. Gently lower the cylinder into the water until fully submerged, without touching the sides.
2. Collect the displaced water in the beaker.

4. Measure weight in water

1. Record the new reading on the spring balance, W_{water}

5. Measure displaced water

1. Pour the collected water into the measuring cylinder.
2. Record the volume, V_{water}



- Calculations

- Buoyant Force from Weight Difference:

$$F_b = W_{\text{air}} - W_{\text{water}}$$

- Weight of Displaced Water:

$$W_{\text{displaced}} = \rho_{\text{water}} \times g \times V_{\text{water}}$$

(Use $\rho_{\text{water}} \approx 1000 \text{ kg/m}^3$, $g \approx 9.81 \text{ m/s}^2$)

- Compare F_b and $W_{\text{displaced}}$. They should be approximately equal.
- Fill in the observations table

Trial	W_{air} (N)	W_{water} (N)	F_b (N)	V_{water} (mL)	$W_{\text{displaced}}$ (N)
1					
2					

Case: Before the session (Demonstrator)

- Perform the experiment in advance
- Ensure equipment is available
 - Balances calibrated
 - Cylinders and beaker not cracked
 - Enough water
- Update the lab manual with clear step-by-step instructions
- Remind students to prepare for the lab session

Case: Before the session (Student)

- Read a pre-lab checklist: Density = Mass / Volume. $W_{\text{displaced}} = \rho_{\text{water}} \times g \times V_{\text{water}}$
- Watch short videos on Archimedes' principle
 - <https://youtu.be/0v86Yk14rf8> and https://youtu.be/_p-hwElkrlk
- Bring necessary lab gear (lab coat, closed shoes, tied hair).
- Search and share real-life uses of density
 - e.g., why ships float, why ice floats
- Answer a self-assessment quiz
 - “What happens to the density if you double the mass but also double the volume?”
 - “If an object floats, what does that tell you about its density compared to water?”

Case: During the session

- Start with a pre-lab talk:
 - Why density matters (e.g., materials science, buoyancy)
 - How to read the meniscus correctly
 - Safety note: handle glassware carefully, mop up spills
- Walk around and check student progress:
 - Are they weighing the object properly?
 - Are they reading the meniscus at eye level?
 - Are they recording results systematically?



Case: During the session

- Position yourself in a central and visible spot.
- Move between tables to check student technique and progress.
- When asked for help, prompt students to explain what they have done so far before giving hints.
 - Avoid doing the experiment for them
 - Ask guiding questions
 - “What would happen if we used a floating object instead?”
 - “How can you reduce human error when reading the cylinder?”
 - Pause to let students think
 - Predict–Observe–Explain
- Encourage peer-to-peer discussion if one group struggles.

Common questions asked by students

- Procedural and observational
 - "How do I perform this step?"
 - "What should I be seeing or measuring right now?"
 - "What counts as a valid observation?"
 - "How do I record the data?"
- Equipment and materials
 - "What does this machine or tool do?"
 - "What materials do I need?"
 - "How do I set up or calibrate this equipment?"
- Conceptual and connection
 - "What does this result mean?"
- Logistics
 - "Can you check our progress?"

Prepare answers for
common questions

Common questions that we can ask students

- Observational and descriptive
 - "What do you observe happening right now?"
 - "What assumptions are being made in this setup?"
 - "What does this machine or tool do, and how are you using it?"
 - "Describe the differences or similarities between these samples/results."
- Predictive and hypothetical
 - "What do you think will happen if we change this variable?"
- Explanatory and reasoning questions
 - "Why do you think this result occurred?"
- Connective and application
 - "How does this experiment connect to what we've covered in lectures or real-life applications?"
- Reflective and evaluative
 - "What would you do differently next time, and why?"
 - "Are there other options available to you for approaching this problem?"

Handling Challenges

- Improper handling of hazardous materials
- Non-compliance with safety protocols
- Malfunctioning equipment
- Unengaged students
 - Ask probing questions, assign roles in groups
- Experiments taking longer than expected
 - Create timelines
- Explaining complex concepts clearly
 - Review materials in advance
- Grading subjective elements (e.g., lab techniques)
 - Use rubrics for consistency

Assessment and Feedback

- Read the checkpoint of the lab session
 - Identify common mistakes during the pre-lab
 - Provide timely and specific feedback
- Read rubrics and checklists if available for marking lab reports
 - Introduction: Aims and background
 - Methods
 - Results: Tables and figures
 - Discussions:
 - Interprets results in relation to the hypothesis; Explains unexpected outcomes
 - What surprised them, what they learned, and what questions remain
 - Conclusions: Reflects on limitations

References

- [Facilitating Laboratory Sections - GSI Teaching & Resource Center](#)